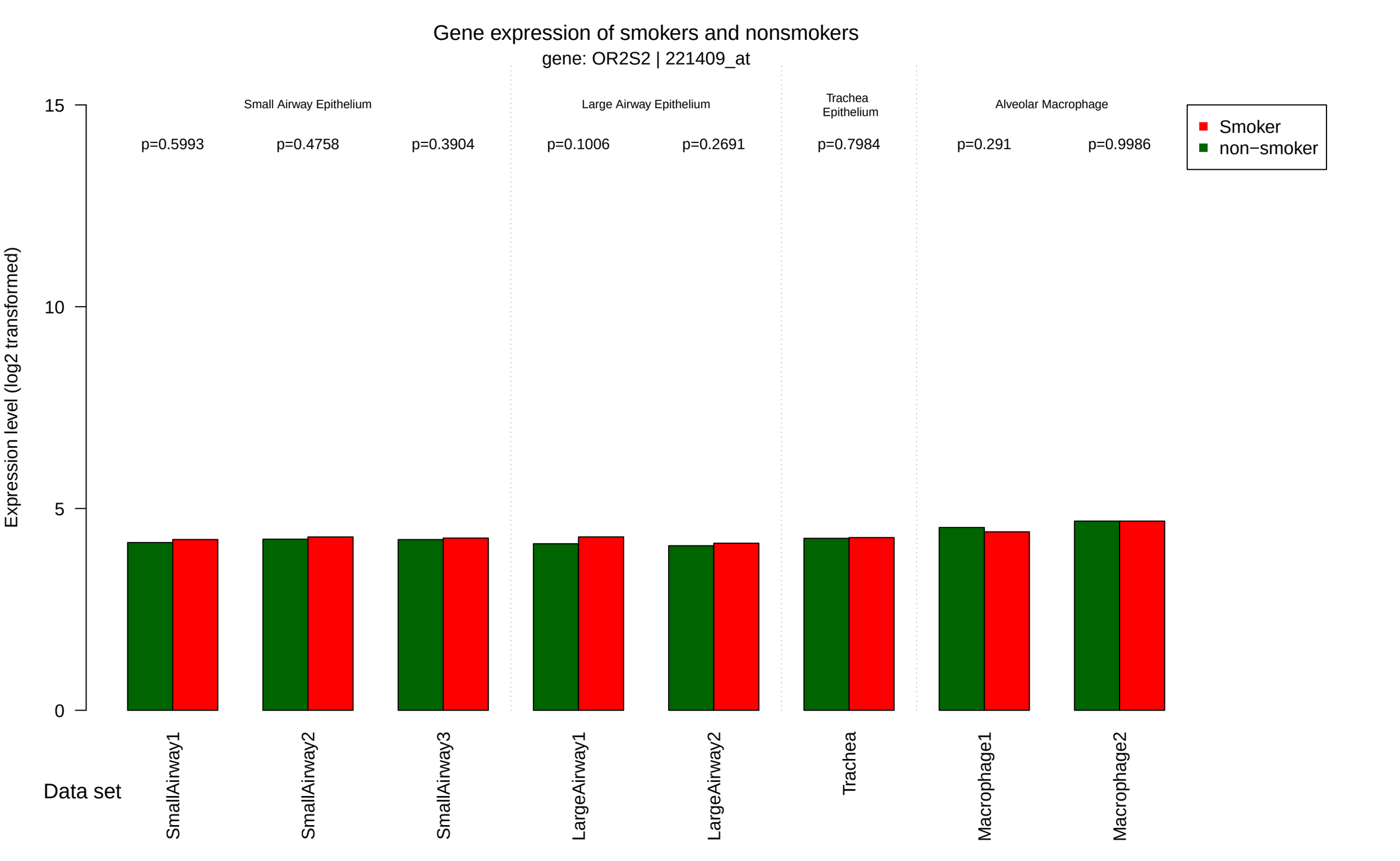
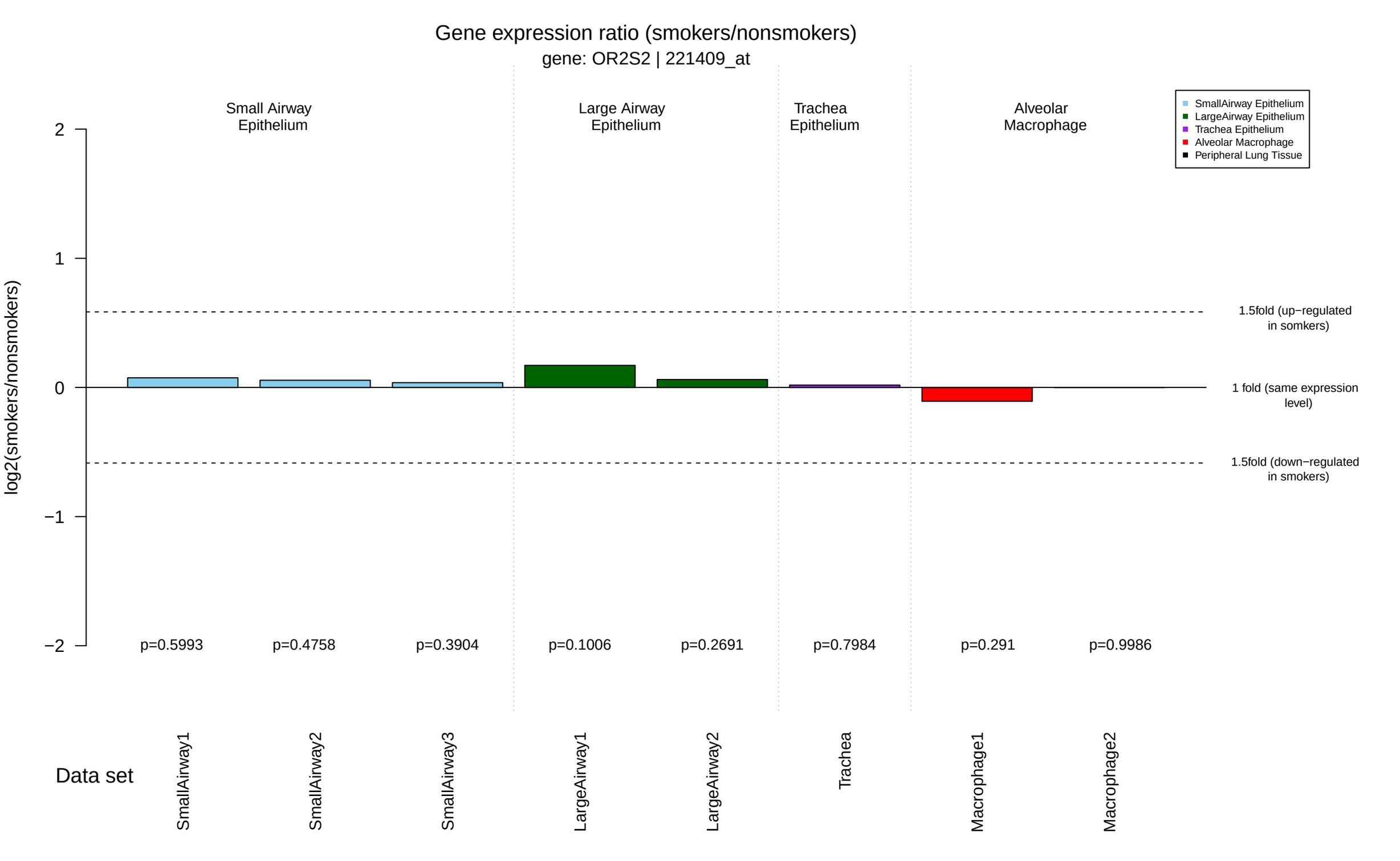
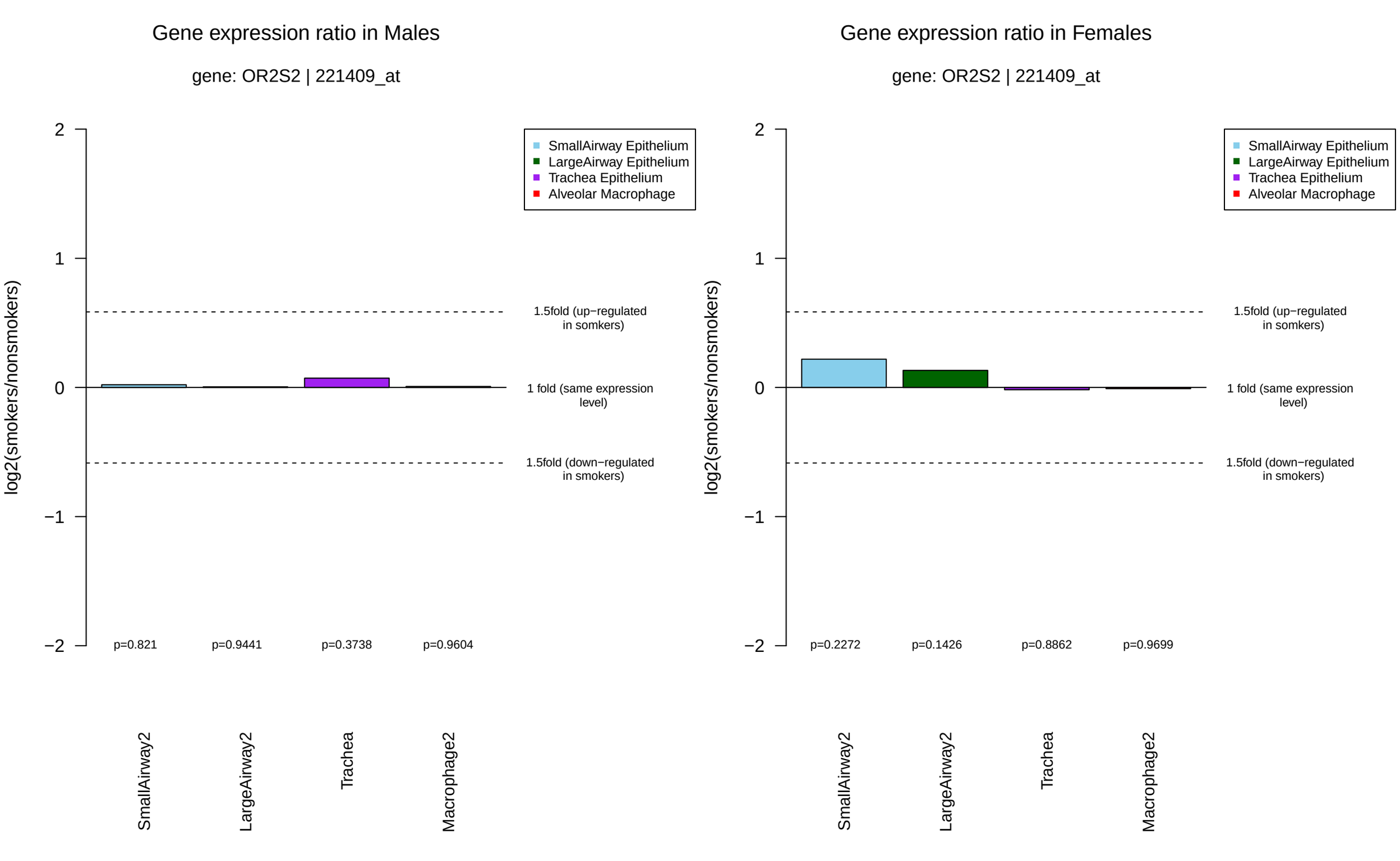
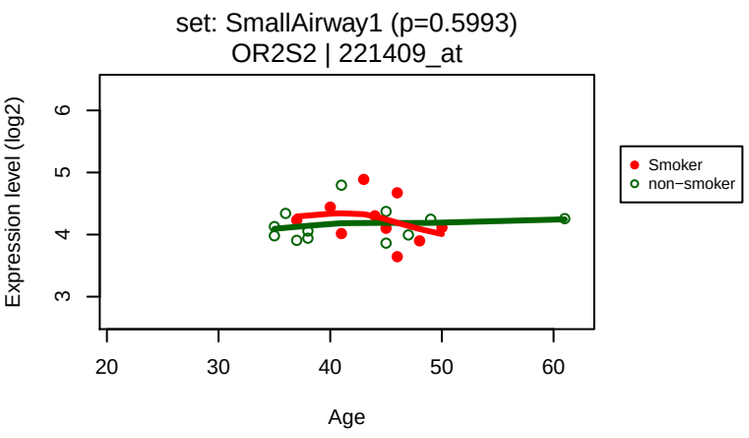
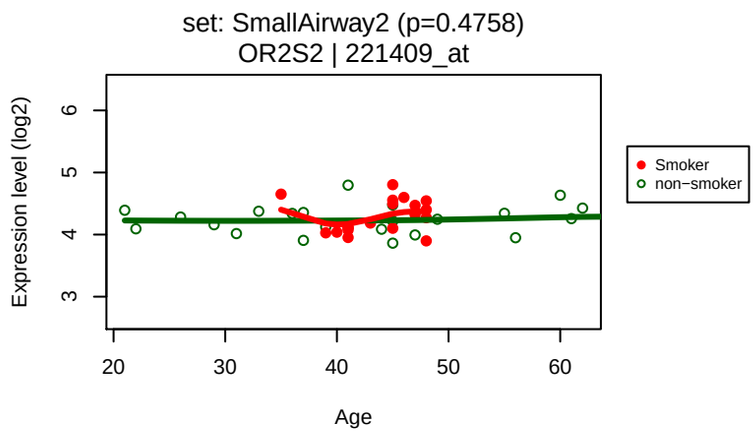
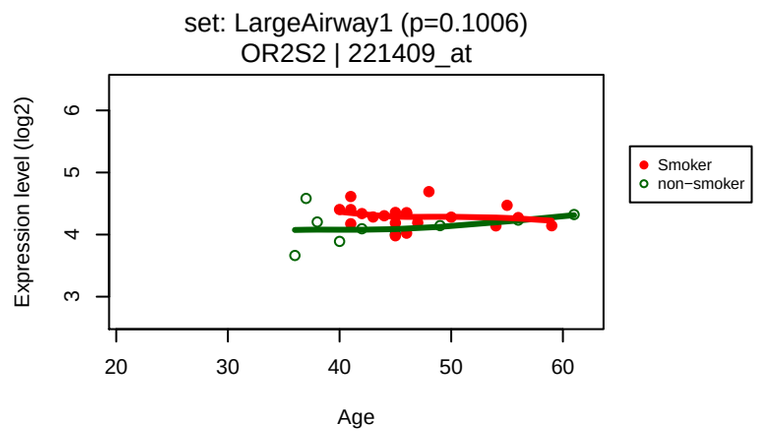

Gene expression of smokers and nonsmokers
gene: OR2S2 | 221409_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5993
p=0.4758
p=0.3904
p=0.1006
p=0.2691
p=0.7984
p=0.291
p=0.9986
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

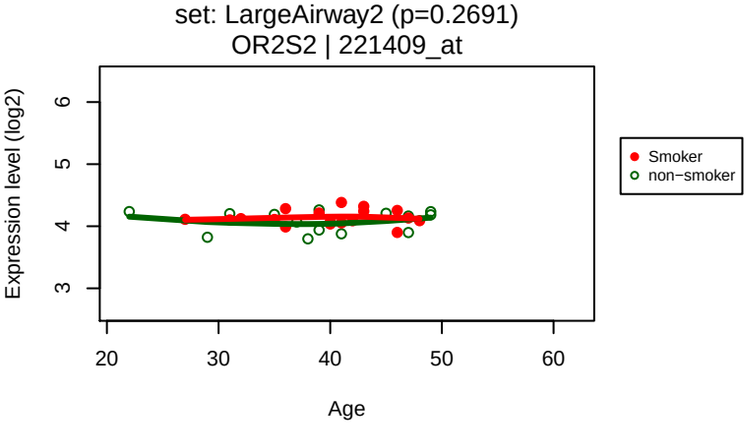

Gene expression ratio (smokers/nonsmokers)
gene: OR2S2 | 221409_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5993
p=0.4758
p=0.3904
p=0.1006
p=0.2691
p=0.7984
p=0.291
p=0.9986
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

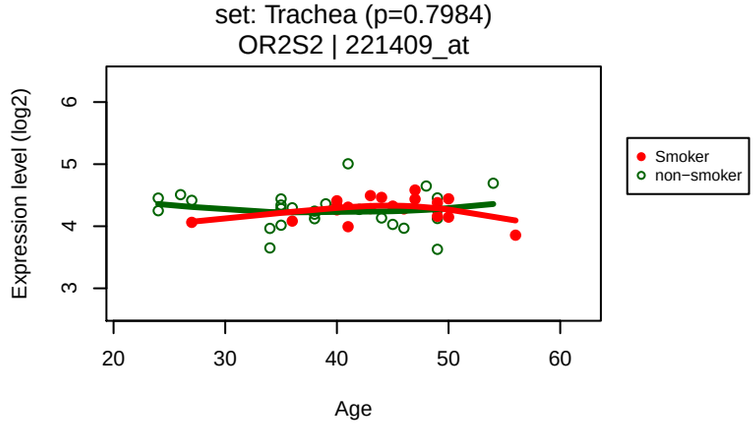

Gene expression ratio in Males
gene: OR2S2 | 221409_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.821
p=0.9441
p=0.3738
p=0.9604
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR2S2 | 221409_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2272
p=0.1426
p=0.8862
p=0.9699
SmallAirway2
LargeAirway2
Trachea
Macrophage2

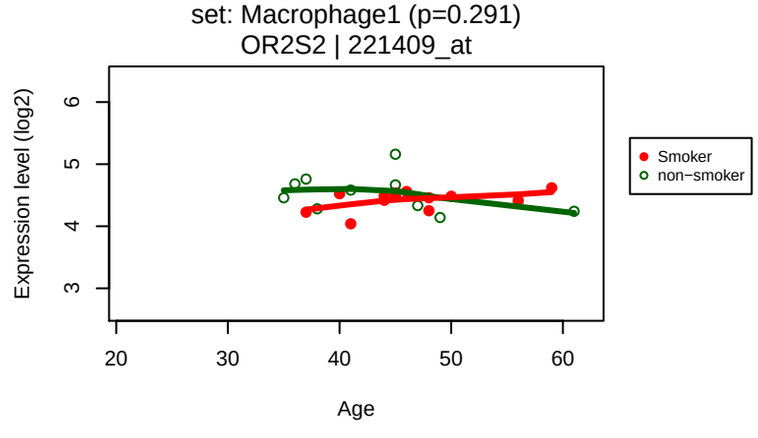

set: SmallAirway1 (p=0.5993)
OR2S2 | 221409_at
Expression level (log2)
Age
Smoker
non−smoker

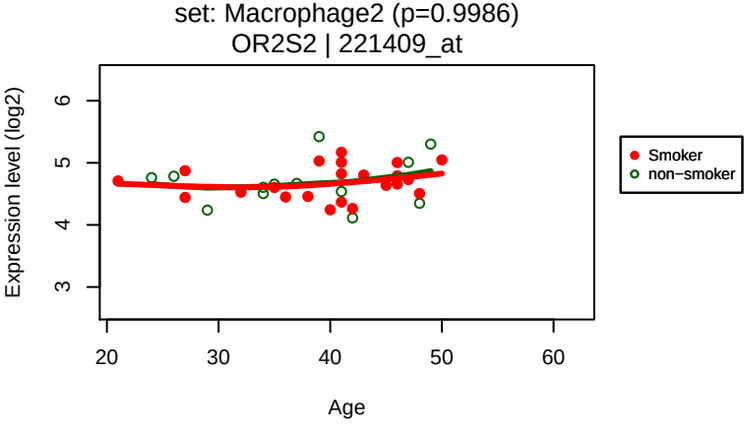

set: SmallAirway2 (p=0.4758)
OR2S2 | 221409_at
Expression level (log2)
Age
Smoker
non−smoker

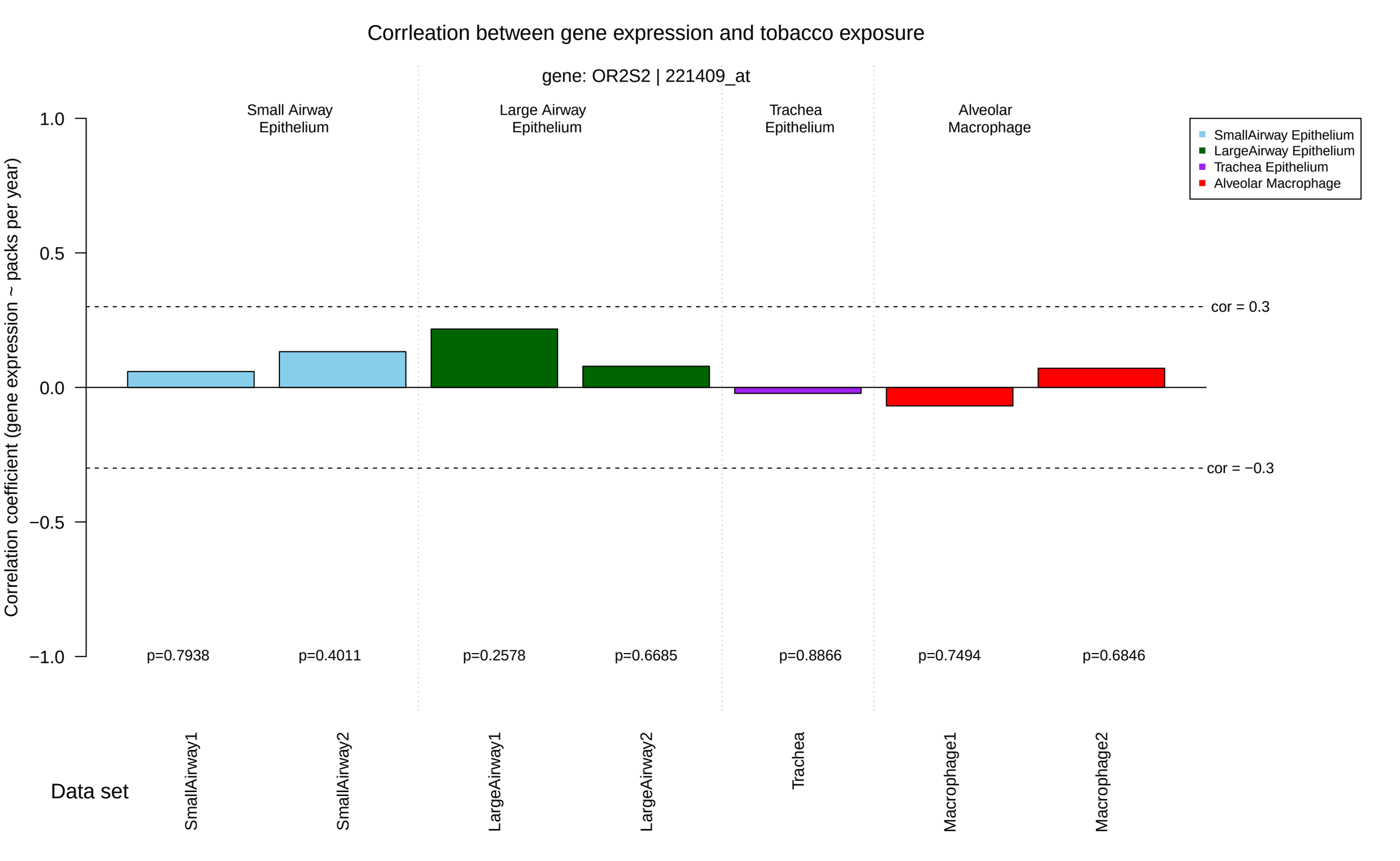

set: LargeAirway1 (p=0.1006)
OR2S2 | 221409_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2691)
OR2S2 | 221409_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7984)
OR2S2 | 221409_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.291)
OR2S2 | 221409_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9986)
OR2S2 | 221409_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR2S2 | 221409_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7938
p=0.4011
p=0.2578
p=0.6685
p=0.8866
p=0.7494
p=0.6846
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2